

THE INTERNET OF THINGS



News, views and development on the Internet of Things

“Cool Tech for ‘Kool’ IoT Products”

The Internet of Things (IoT) is top-most on most electronics companies' agenda today. Texas Instruments (TI) is no different. Their wide connectivity portfolio is powering several interesting IoT products and platforms across the globe, such as the 'Internet of KoolThings' platform created by DeviceLab Inc. and KoolTechs Inc. Praveen Ganapathy, director, business development, Texas Instruments (India), speaks to Janani Gopalakrishnan Vikram about what's trending in this space

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Which is the most exciting 'TI-powered' IoT product according to you? What does it do and what value does TI add to it?

One of the most exciting products that TI is powering is the Bluetooth low energy based 'Internet of KoolThings' platform. This platform involves an expandable multi-sensor, multi-function system that packs features found in many different consumer product categories into a user-friendly 'iWanditAll' KoolThings platform. iWanditAll KoolThings includes KoolWand (a wearable key fob-like device) and KoolBridge (a small portable wireless hub/media player), and is expandable with optional KoolSensors and KoolAccessories.

TI's SimpleLink CC254x Bluetooth low energy wireless MCUs and TI's power, battery management solutions presently power the KoolWand. Additionally, the KoolBridge employs an optional RF range extender chip from TI (CC2591) to improve the covered range.

In this manner, TI's versatile connectivity portfolio is helping customers connect more to the IoT. In addition to the embedded wireless connectivity technologies, customers can benefit from the industry's leading portfolio

of microcontrollers, processors and analogue solutions for their complete application.

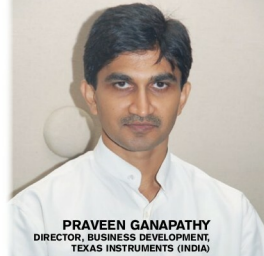
What do you think is the topmost security concern when it comes to IoT products, and how do you address it in your chips, firmware and platform?

One of the biggest security concerns for the IoT ecosystem is the breach of personal data. As the IoT involves several devices, security is vital at all layers. To avoid these circumstances, several players in the IoT are opting for hardware with security technology.

As part of our embedded and connectivity technology portfolio we have security solutions to prevent, detect and respond to unintended or malicious behaviour. They address the hardware, software and connectivity spaces in the IoT through symmetric cryptography, authentication and anti-cloning, IP protection and tamper protection.

As a tech provider, can you tell us which wireless technologies are most product makers seeking to imbibe in their products? What could be the reason?

We are seeing the popularity of all low-power radio technologies being imbibed into IoT products. These are a critical aspect of the IoT because all the devices must continuously or periodically be connected. One size does



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not fit all low-power radio technologies. So, depending on the product, engineers can adopt a suitable low-power radio technology based on their capabilities. At TI, we are delivering solutions that are compatible with ZigBee, Wi-Fi, Bluetooth, Sub-1GHz, EtherCAT, Prime Alliance, G3-PLC Alliance and others. As mentioned, based on the end application at hand, IoT players can incorporate the best suited connectivity solution for their design, whether it is Bluetooth, Wi-Fi or EtherCAT, etc.

Anything new and exciting that TI is working on for its IoT customers?

One of our most exciting product lines, already available in the market, is TI's SimpleLink solution. This range includes smart RF transceivers, wireless networking solutions, wireless MCUs and range extenders. Each product is supported with its own design ecosystem starting with evaluation and development tools to reference designs, training and extensive online support tools. This enables you to create your wireless solution with TI helping you from start to finish. ●

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